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IN-SITU MYCOREMEDIATION SYSTEM AND PROCESS

Abstract

An in-situ mycoremediation system and process is provided, including a device with a rod casing having a top end, a bottom end, and a sidewall with one or more perforations, the sidewall defining an internal channel that extends from an intake opening on the top end to the one or more perforations, a sleeve that extends around at least part of the rod casing and that is slidable between at least a first position that covers the one or more perforations and a second position that at least partly uncovers the one or more perforations, and a plumbing line linked to the intake opening and configured to facilitate forcible injection of one more fungal mixtures and/or air via the one or more perforations when the sleeve is in the second position.

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Background/Summary

PRIORITY CLAIM [0001] This application is a continuation of U.S. non-provisional patent application Ser. No. 17/856,402 filed Jul. 1, 2022, titled IN-SITU MYCOREMEDIATION SYSTEM AND PROCESS (DOCKET GD-P-03-00-NP-01-0000), which application is a non-provisional patent application of U.S. provisional patent application 63/217,756 filed Jul. 1, 2021, titled SYSTEM AND PROCESS FOR MYCOREMEDIATION (DOCKET GD-P-03). [0002] This application claims the benefit of and/or priority to each of the foregoing patent applications and any and all parent, grandparent, and great-grandparent applications thereof. The foregoing patent applications are incorporated by reference in their entirety as if fully set forth herein.

FIELD OF INVENTION
[0003] Embodiments of the invention relate generally to contamination remediation and, more specifically, to an in-situ mycoremediation system and process.
BACKGROUND
[0004] Heavy metals and toxic chemicals and pollutants from human activity accumulate in our soil and water. These contaminants can damage the environment, end up in our food chain, and bioaccumulate in humans, contributing to a wide array of development and health issues.
[0005] Mycoremediation is a method that utilizes fungi mycelium as a remedial treatment to restore balance in soil and water. Mycelium refers to the ultra-fine and dense network of branching thread-like white hyphae that is the vegetative part of the fungi. As mycelium spreads, it secrets enzymes including peroxidases, liginases, cellulases, pectinases, xylanases, and oxidases. These enzymes break down and remove a wide array of contaminants, including heavy metals, organic pollutants, textile dyes, leather tanning chemicals, wastewater, petroleum fuels, polycyclic aromatic hydrocarbons, pharmaceuticals, personal care products, pesticides, herbicides, and others. Furthermore, fungi mycelium can also act as a filter to extract and hyperaccumulate contaminants.
[0006] There are two primary techniques of mycoremediation: ex-situ and in-situ. Ex situ techniques involve excavating contaminants from polluted sites and subsequently transporting them to another site for treatment. In-situ techniques involve treating contaminants at the site of pollution. Ex-situ mycoremediation tends to be expensive due in part to additional costs attributed to excavation and transportation. Ideally, in-situ techniques ought to be less expensive and more efficient as compared to ex-situ mycoremediation techniques, but challenges of inoculating subsurface locations with fungi mycelium, especially soil with low permeability or deeper contamination, has been an obstacle.
SUMMARY
[0007] This disclosure relates generally to contamination remediation and, more specifically, to an in-situ mycoremediation system and process.
[0008] In one embodiment, a device for in-situ mycoremediation includes, but is not limited to, a rod casing having a top end, a bottom end, and a sidewall with one or more perforations, the sidewall defining an internal channel that extends from an intake opening on the top end to the one or more perforations; a sleeve that extends around at least part of the rod casing and that is slidable between at least a first position that covers the one or more perforations and a second position that at least partly uncovers the one or more perforations; and a plumbing line linked to the intake opening and configured to facilitate forcible injection of one more fungal mixtures and/or air via

the one or more perforations when the sleeve is in the second position.

[0009] In another embodiment, a system for in-situ mycoremediation includes, but is not limited to, a set of rod casings each having one or more perforations, the set of rod casings configured to be distributedly driven into a subsurface zone; a pump; and one or more plumbing lines that link the pump to the set of rod casings to facilitate forcible injection of one more fungal mixtures and/or air via the one or more perforations into the subsurface zone.

[0010] In a further embodiment, a process for in-situ mycoremediation includes, but is not limited to, driving one or more rod casings into a subsurface zone, the one or more rod casings including one or more orifices that are covered; partially retracting the one or more rod casings to expose the one or more orifices; forcible injecting a fluid via the one or more orifices to initiate fracture in the subsurface zone; forcibly injecting a fungal nutrient mixture via the one or more orifices into the subsurface zone; and ventilating the subsurface zone with air via the one or more orifices to support a hyphal network of mycelium in the subsurface zone.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0011] FIG. **1** is an environmental view of an injection remediation system, in accordance with an embodiment;

[0012] FIG. **2** is a component diagram of an injection remediation system, in accordance with an embodiment;

[0013] FIGS. **3-4** are perspective views of a device for mycoremediation, in accordance with an embodiment of the invention;

[0014] FIG. **5** is perspective view of a device for mycoremediation, in accordance with an embodiment of the invention;

[0015] FIGS. **6-8** are environmental perspective views of a device for mycoremediation, in accordance with an embodiment of the invention; and

[0016] FIG. **9** is a flow diagram of a process for mycoremediation, in accordance with an embodiment of the invention.

DETAILED DESCRIPTION

[0017] This disclosure relates generally to contamination remediation and, more specifically, to in-situ mycoremediation. In one embodiment, a system, device, and process are provided to use fungi to alter subsurface conditions in a beneficial manner. By inoculating the soil with one or more species of fungi, different goals can be realized. Certain embodiments are set forth in the following description and in FIGS. **1-9** and APPENDIX A and B provide a thorough understanding of such embodiments. APPENDIX A and B are incorporated by reference fully herein.

[0018] Many pollutants, such as solvents, fuels, PFAS, dyes, pharmaceuticals, herbicides, heavy metals, detergents, cyanotoxins and a large percentage of the world's polycyclic aromatic hydrocarbons (PAHs) exist in marine and terrestrial subsurface zones such as vadose zones, sediment, groundwater and even mangrove systems, including substrates such as dirt, sediment, silty sand, sand, and silt.

[0019] Certain fungal species are usable to degrade pollutants in subsurface environments where contamination has migrated. For example, white-rot fungi (ligninolytic) secrete enzymes to catalyze the oxidation of PAH into quinones, which then further degrade, by ring fission, into CO2. Other species such as *Phanerochaete chrysosporium* and *Pleurotus* ostreatus are also usable. *Pleurotus* ostreatus and *Trametes villosa* and *Coriolopsis* can degrade crude oil. Genera *Aspergillus* and *Botrytis* can degrade agricultural contaminants (i.e., insecticides, herbicides, and livestock hormones).

[0020] Biochemical actions of fungi occur through excretion of organic acids and other metabolites. For example, arbuscular mycorrhizal fungi (AMF) live in symbiotic association with plants through the roots and excrete proteins and acids that alter soil and sediment structure and increase connectivity across soil pores and also increase plant tolerance of saline substrates. Wood-rotting (decomposer) fungi physically influence substrate structure by weaving mycelium through the substrate and chemically by excreting ligninocellulosic enzymes (including lignin peroxidase (LiP), manganese peroxidase (MnP) and laccase) which break down complex carbon-based chains into shorter chains increasing substrate permeability and facilitating biodegradation of organic contaminants by bacteria and other microbes. Anaerobic (or low oxygen needs) yeasts and molds and other fungi have been isolated from the types of environmental conditions where contaminants are found, including saline sediments and aquatic environments with oil spills. Other extremophilic molds and yeasts, including black yeasts, have been isolated from groundwater environments.

[0021] To increase effectiveness of fungal species in degradation of pollutants, in one particular instance, an innovative approach is used to create robust growth of fungi in a hyphal network for mycorenediation. In aquatic or terrestrial environments, soil porosity, permeability, and hydraulic conductivity are first increased by fracturing a subsurface with a first fluid, such as a carbon solution, at a high pressure. Then, a slurry of fungi with a food source is disbursed into the disturbed subsurface under a pressure that is lower than that of the first fluid to initiate and support colonization and/or formation of a matrix of mycelium. The matrix of mycelium itself increases soil porosity and produces extracellular ligninolytic enzymes that breakdown and degrade pollutants. Further, the matrix of mycelium acts to absorb and/or filter various pollutants, such as heavy metals to support remediation. Additionally, the area of influence can be increased via the addition of more sugar solution to encourage the migration of fungus. More solution may be added to the original injection pipe, or via one or more additional injection pipes to encourage hyphal fungal growth. Additionally, either before, after, or during any of the foregoing operations, an optional injection of oxygen or air can be forcibly injected to the subsurface or encouraged through venting. It is also possible to restrict access to air in the subsurface for anoxic substrates.

[0022] Porosity is a measure of the void spaces in a substrate and permeability is a measure of the ability of a substrate to transmit fluids. Poor substrate permeability and porosity can limit microbe's ability to access contaminants in subsurface environments and break them down. An increase in porosity can be accomplished using the high pressure fluid, but certain fungi are known to alter substrate permeability and porosity and act as ‘geologic agents’. Thus, the initial high pressure fluid injection operation may be supplemented or omitted in favor of a fungal slurry pressure injection that itself increases soil porosity and forms a matrix of mycelium to break down pollutants.

[0023] Methods and devices for introducing the fungus into the soil matrix can vary depending on soil conditions. In one embodiment, an injection pipe of about 1.5-inch diameter is driven or vibrated into the ground with a “direct push” type rig, often a Geoprobe. As the zone of interest is reached, the rods are retracted slightly (e.g. 18-inches), which exposes a series of injection slots in the drive pipe. A propagation notch is started using pressurized water. This notch is about 4 to 6-inches deep into the lateral soil horizon. An inoculation fluid is then pumped into the pipe under pressure until the overburden pressure of the soil is exceeded and the horizon fractures. The lateral fracture then extends out horizontally and is approximately 1-inch wide and has approximately a 30-degree radial expression. Thus, the tooling has the capability to inject both aqueous and viscous slurry solutions, such as those including grain/sawdust material. In certain embodiments, this tooling allows for higher hydraulic pressure to help create fractures in low porosity soils and allows for lower hydraulic pressure to support injection of mycelium and/or enzymes at a lower sustained pressure.

[0024] The injection fluid or inoculation slurry can also vary depending on soil conditions and/or pollutants. In one particular embodiment, the inoculation contains a mixture of fungal spores and a sugar solution as a food source. Other slurries are usable for inoculation. For instance, ingredients may include one or more liquids, hagfish slime, wood, sawdust, grain spawn, EVO, and/or carbon. In one particular embodiment, a sawdust slurry is provided with one or more hardwoods and soybean hull. In another particular embodiment, a grain slurry is provided with one or more rye grains. In a further embodiment, a liquid slurry is provided with dextrose and malt.

[0025] Possible mediums delivered for mycoremediation in the subsurface may include mycelium, but optionally or alternatively may include any one or more of the following: liquid mycelium, solid-state mycelium on a carbon source, fungi-inoculated seedlings, and/or encapsulated enzymes. The table in APPENDIX A outlines usable mediums and species for use with various subsurface conditions.

[0026] Various injection equipment can be employed for injecting into a subsurface, including one or more injection pumps. Higher pressure pump(s) are usable to fracture soil prior to injecting mycelium and lower pressure pump(s) are usable for a liquid culture injection of mycelium/enzymes. Additionally, a pressurized atmospheric gas may be employed for pneumatic injection. Optionally, a progressive cavity pump or centrifugal pump can be employed.

[0027] FIG. **1** is an environmental view of an injection remediation system for injecting one or more liquids, slurries, and/or mixtures as illustrated and described herein, in accordance with an embodiment. In one embodiment, a system **100** for in-situ mycoremediation includes, but is not limited to, a set of rod casings **112** each having one or more perforations, the set of rod casings **112** configured to be distributedly driven into a subsurface zone **101**; a pump **108**; and one or more plumbing lines **114** that link the pump **108** to the set of rod casings **112** to facilitate forcible injection of one more fungal mixtures and/or air via the one or more perforations into the subsurface zone **101**.

[0028] An injection remediation system **100** includes, but is not limited to, at least one reservoir **102***a* configured to contain material; at least one load sensor **116***a* supporting the at least one reservoir **102***a*; at least one dosing pump **106***a* operably coupled to the at least one reservoir **102***a* and configured to controllably source the material. In one particular embodiment, one or more plumbing lines **114** connect the at least one reservoir **102***a* to at least one injection pipe **112**. In a further embodiment, at least one injection pump **108** is associated with the one or more plumbing lines **114** and is configured to pressurize the material. In a further embodiment, at least one manifold **110** includes one or more auxiliary ports configured to distribute the material. In yet another embodiment, the system **100** further includes at least one additional reservoir **102***b* configured to contain the same or different material; at least one additional load sensor **116***b* supporting the at least on additional reservoir **102***b*; and at least one additional dosing pump **106***b* operably coupled to the at least one additional reservoir **102***b* and configured to controllably source the different material. In certain embodiments, the system **100** is incorporated with vehicle **118**.

[0029] In certain embodiments, injection system **100** provides or conducts in situ remediation of contaminant plumes in terrestrial or marine surfaces or subsurfaces. Using portable drilling and direct push machines, such as with a Geoprobe 7822DT/3230/420M, or Ditch Witch R300, injection pipes are driven or pushed into the subsurface. At various depths, the subsurface is fractured using compressed gas or pressurized fluid to initiate fluid or material flow into the target soil horizon. Depending on the contaminant of concern, various oxygenators, air or oxygen gas, and/or fungal mixtures are introduced into these fractures to facilitate mycoremediation. The various injection fluids or materials can be stored, mixed, or delivered using a portable injection platform. The platform, which can be truck, trailer, or vessel mounted, contains various raw material tanks, mix tanks, injection pumps, and/or control hardware and/or software.

[0030] In one embodiment, the vehicle **118** includes any car, truck, van, trailer, rig, ship, boat, underwater device, or any other manned or unnamed machine. The vehicle **118** as depicted includes a truck and trailer combination with an enclosed area for housing the injection remediation components. Within the trailer, the vehicle **118** includes a mixing tank/reservoir **102***a*, mixing tank/reservoir **102***b*, raw material tank **104***a*, and/or raw material tank **104***b* configured to source and/or mix injection materials or fluid. The dosing pump **106***a* and/or dosing pump **106***b* controllably source the injection materials using plumbing lines **114** connected to the mixing tank/reservoir **102***a*, mixing tank/reservoir **102***b*, raw material tank **104***a*, and/or raw material tank **104***b*. The sourced injection material is fed via the plumbing lines to an injection pump **108** that pressurizes and the injection material. The manifold **110** includes a plurality of output ports and distributes the injection material to a plurality of injection pipes **112** disposed in an injection mycoremediation area **101**. Load sensor **116***a* and/or load sensor **116***b* are disposed under the mixing tank/reservoir **102***a* and/or mixing tank/reservoir **102***b* and are configured to provide weight output, respectively.

[0031] In certain embodiments, any components of the system **100** can be omitted, augmented, or differently arranged. For instance, either of the mixing tank/reservoir **102***a* and the mixing tank/reservoir **102***b* can be omitted or augmented to provide no mixing tanks and/or reservoirs, a single mixing tank and/or reservoir, or a plurality of mixing tanks and/or reservoir. Likewise, any of the raw material tank **104***a* and raw material tank **104***b* can be omitted or augmented to provide no raw material tanks, a single raw material tank, or a plurality of raw material tanks. Any mixing tanks and/or reservoir, such as mixing tank/reservoir **102***a*, can include zero, one, or a plurality of connected raw material tanks, such as raw material tank **104***b*, that are configured to source raw material ingredients. Similarly, any of the dosing pump **106***a* and dosing pump **106***b* can be omitted or augmented to provide zero, one, or a plurality of dosing pumps. Additionally, dosing pumps can be provided to source raw material from any of the raw material tanks, such as raw material tank **104***b*, for any mixing tank/reservoir, such as mixing tank/reservoir **102***a*. Thus, a dosing pump, such as dosing pump **106***b*, can be positioned to source raw materials from a raw material tank and/or a mixing tank and/or reservoir. Additionally, a dosing pump, such as dosing pump **106***a*, can be positioned to transfer mixed material from one mixing tank/reservoir to another mixing tank/reservoir, such as from mixing tank/reservoir **102***a* to mixing tank/reservoir **102***b*. In other embodiments, the injection pump **108** can be augmented or differently positioned. For instance, a plurality of independent injection pumps can be provided for each mixing tank/reservoir, such as mixing tank/reservoir **102***a*, and/or each raw material tank, such as raw material tank **104**. The injection pumps can also be combined in parallel and/or in series, and can be disposed downstream of any manifold, such as manifold **110**. In further embodiments, the plumbing lines **114** are different configured. For example, plumbing lines **114** can connect each of the mixing tank/reservoir **102***a* and mixing tank/reservoir **102***b* independently to the injection pump **108** for combination prior to the manifold **110** as illustrated. Alternatively, plumbing lines can connect each of the mixing tank/reservoir **102***a* and mixing tank/reservoir **102***b* to independent injection pumps and/or manifolds to join downstream, such as at or proximate to the injection pipe **112**. Likewise, plumbing lines **114** can connect a raw material tank, such as raw material tank **104***a*, directly to an injection pump **108** or downstream of a manifold, such as manifold **110**, thereby bypassing a mixing tank and/or reservoir, such as mixing tank/reservoir **102***a*. The components of system **100** and their arrangement are exemplary, but many modifications can be made to accomplish a particular injection mycoremediation mission.

[0032] In some embodiments, the raw material tank **104***a* and/or raw material tank **104***b* are containers or reservoirs that are configured to store and/or source ingredients for injection mycoremediation. The raw material tank **104***a* and/or raw material tank **104***b* are approximately to approximately 300 gallons, but other sizes are possible. The raw material tank **104***a* and/or raw material tank **104***b* are composed of steel, aluminum, carbon fiber, plastic, or other metal, synthetic, or composite material. The injection materials stored can include a variety of liquids, solids, or mixtures, including any of fungus, mycelium, wood chips, enzymes, carbon, bacteria, microbes, fiber, water, oil, polymers, iron filings, vegetable oil, persulfate, hydroxide, or other ingredients disclosed herein or equivalent thereto. Using one or more plumbing lines **114**, gravity, a feeder, a pump, a duct, or other dispenser, the raw material tank **104***a* and/or raw material tank **104***b* can source ingredients to one or more of the mixing tank/reservoir **102***a* and/or mixing tank/reservoir **102***b*. Alternatively, the raw material tanks **104***a* and/or raw material tank **104***b* can source material directly to an injection pump **108** without first entering a mixing tank/reservoir, such as mixing tank/reservoir **102***a* or mixing tank/reservoir **102***b*. Additionally, either of the raw material tanks **104***a* and/or raw material tank **104***b* can source material jointly to one or more mixing tanks/reservoir, such as mixing tank/reservoir **102***a*. One or more dosing pumps can control an ordered sequence of sourcing from one or more of the raw material tanks **104***a* and/or raw material tank **104***b*.

[0033] In additional embodiments, the mixing tank/reservoir **102***a* and/or the mixing tank/reservoir **102***b* are configured to mix injection materials. The mixing tank/reservoir **102***a* and/or the mixing tank/reservoir **102***b* can be approximately 5 to 300 gallons, but other sizes are possible. The mixing tank/reservoir **102***a* and/or the mixing tank/reservoir **102***b* can be composed of steel, aluminum, carbon fiber, plastic, or other metal, synthetic, or composite material. Either of the mixing tank/reservoir **102***a* and/or the mixing tank/reservoir **102***b* include one or more dry material feeders, one or more dry material eductor, one or more liquid dosing manifolds, one or more bulk dry feed systems, one or more fungal sources, and/or connections to one or more raw material tanks or reservoirs, such as raw material tank **104***b*. The mixing tank/reservoir **102***a* and/or the mixing tank/reservoir **102***b* can be equipped with a slight conical base, such as with an approximately 1″ to approximately 3″ slope terminating at a drain plug. Internal to the mixing tank/reservoir **102***a* and/or mixing tank/reservoir **102***b* includes one or more splash fins and/or a mixer. A mixer motor and/or gear box can be coupled to the mixer of the mixing tank/reservoir **102***a* and/or the mixing tank **102***b*, such as a 0.5 to 5 hp electric or gas motor at approximately 1000 RPM to approximately 5000 RPM. A lid of the mixing tank/reservoir **102***a* and/or mixing tank/reservoir **102***b* includes a hinge with a friction or other locking mechanism that exposes one or more couplers, such as 1″ or 2″ full couplers. The mixing tank/reservoir **102***a* and/or the mixing tank/reservoir **102***b* can independently mix the same or different injection materials simultaneously or in series. For instance, to maintain continuity and minimize disruption of injection remediation, the mixing tank/reservoir **102***a* can be dispensing an injection material formerly mixed, while the mixing tank/reservoir **102***b* is preparing an additional batch of the same injection material for subsequent dispensation. Alternatively, to isolate injection materials that could chemically react or that are designed to inject at different pressure, the mixing tank **102***a* can be dispensing a first injection material of fungus, for example, while the mixing tank **102***b* can simultaneously be dispensing a second injection material of fluid, whereby the first and second injection materials are sequentially injected via one or more injection heads **112** using dual manifolds **110**, for example.

[0034] In one embodiment, one or more load sensors, such as load sensor **116***a*, are disposed below or otherwise support any one or more of a mixing tank, reservoir, or raw material tank. The one or more load sensors are configured to convert downward force exerted by mass or weight into one or more electrical signals. The resultant output of the one or more load sensors is used to determine a weight.

[0035] In further embodiments, the system includes one or more dosing pumps, such as dosing pump **106***a* and dosing pump **106***b*. Each of the mixing tanks/reservoir **102***a* and mixing tank/reservoir **102***b* include a dosing pump, but also any of the raw material tanks **104***a* and **104***b* may be associated with a respective dosing pump. Certain ones of the reservoirs, raw material tanks, or mixing tanks can include zero, one, or a plurality of dosing pumps. Additionally, any of the dosing pumps can operate to transfer material to or between one or more reservoirs, raw material tanks, or mixing tanks. The dosing pumps can further be used to source injection material from any of the reservoirs, raw material tanks, or mixing tanks and deliver the same to a mixture line, an injection pump, a manifold, an injection head, and/or other plumbing.

[0036] In further embodiments, the system **100** includes an injection pump **108**, which can include a D35 hydraulic injection pump, a 3L8 hydraulic injection pump, a centrifugal-type electric injection pump, or any other pressurizing pump. The injection pump **108** operates to pressurize the injection material or injection material mixture and force the injection material through the one or more injection pipe **112**. A plurality of injection pumps **108** may be used, such as with a plurality of or dual manifolds **110** that operate to independently deliver respective injection materials or injection material mixtures at the injection mycoremediation site **101**. The dual or plurality of manifold **110** options are usable in contexts where chemical reactions may take place between injection materials and/or where injection material isolation is desired until a particular point in the system **100** or at the site **101**, such as when differential pressure is desired or required between different injection materials. Alternatively, a series of injection pumps **108** can operate over lengths of plumbing to repressurize injection material due to pressure loss along a length of plumbing **114** or to further boost pressure of injection material. In certain embodiments, the injection pipes **112** comprise a plurality of injection pipes extending from a multi-ported manifold **110**. For instance, the manifold **110** can include approximately 2 to 30 auxiliary ports each extending via flexible lines such as wire braided high-pressure-rated flexible tubing to respective injection pipes **112**. The injection pipes **112** are positioned approximately 2 to 30 feet apart to collectively impact a subsurface zone at the injection remediation site **101**. Other configurations of injection pipes **112** are within the scope of the present disclosure.

[0037] For example, system **100** can store species of Pearl Oyster, Turkey Tail, Phoenix Oyster, King Stropharia, or other fungus or mycelium; media of water, liquid, sawdust, seeds, grass, enzymes, carbon, sugar; and/or gases or gaseous solutions in one or more of the reservoirs **102**. The one or more dosing pumps **106** are usable to transfer any of the stored ingredients between reservoirs **102**, such as for mixing to create inoculation mixtures. Such inoculation mixtures may include, for example, a liquid culture of aerobic fungal decomposers, myceliated sawdust of aerobic fungal decomposers, inoculated seeds, encapsulated enzymes, or the like. The inoculation mixture in one or more reservoirs **102** is deliverable under pressure using the one or more injection pump **108** via the one or more plumbing lines **114** via the one or more injection pipes **112** into the mycoremediation site **101**. A fracture media stored in a separate reservoir **102** can be delivered under higher pressure prior to the inoculation mixture, such as using one or more injection pumps **108**. The injection pump **108** can be variably adjusted to one or more different pressure settings or a plurality of different injection pumps **108** can be employed to generate different pressures for the inoculation mixture and/or fracture media.

[0038] FIG. **2** is a component diagram of an injection remediation system, in accordance with an embodiment. In one embodiment, a injection remediation system **200** includes, but is not limited to, a first reservoir **206** configured to contain first material, the first reservoir including a first load sensor **208** and a first dosing pump **204** for sourcing the first material; at least one injection pipe **218**; one or more plumbing lines **220** connecting the first reservoir **206** to the at least one injection pipe **218**; at least one processor **202** configured to control the first dosing pump **204** to maintain or establish a specified flow volume, mass, rate, or ratio associated with the first material. The processor **202** is electrically coupled via one or more conductors **222** to any of the load sensor **208**, the level sensor **224**, and/or the dosing pump **204**. Optionally, the reservoir **206** includes a level sensor **224** configured to output information associated with a volume or level of the first material within the reservoir **206**. An injection pump pressurizes the first material for distribution via the manifold **216**.

[0039] The system **200** can include a different arrangement of any of the components in order to meet the requirements of one or more injection mycoremediation missions. For instance, the processor **202** can comprise a plurality of processors and/or can include electronics, circuitry, integrated circuits, memory, software, or other computer modules configured to implement operations disclosed herein. The reservoir **206** can comprise one or more reservoirs, holding tanks, dispensers, mixers, or other containers, including a plurality of raw material tanks and/or mixing tanks configured to store and/or mix one or more injection materials, such as fungus, mycelium, sawdust, fiber, carbon, enzymes, microbes, liquid, dissolved gas solution, or any other ingredient disclosed herein or equivalent thereto. Any of the one or more reservoirs **206** can include a load sensor **208** and/or a level sensor **224**. Thus, in an instance where two reservoirs **206** are provided, each of the two reservoirs **206** can include an independent load sensor **208** and/or an independent level sensor **224**. A dosing pump **204** can also be provided for each of the one or more reservoirs to enable independent dosing and/or metering of injection material from the respective one or more reservoirs **206**. The dosing pump **204** can be configured to support flow of injection material from one or more reservoirs **206** to the injection pump **214** or, alternatively, can be configured to support flow of injection material from one or more reservoirs **206** into one or more other reservoirs **206**, such as to mix injection materials together. While one injection pump **214** has been depicted, it is within the scope of the present disclosure to include two or more injection pumps **214**. Likewise, a plurality of manifolds **216** can be provided to distribute injection material to one or more injection pipes **218**. Therefore, in certain embodiments, the plumbing lines **220** are configured to combine one or injection materials prior to the injection pump **214** for distribution via the manifold **216**. And, in other embodiments, the plumbing lines **220** are configured to isolate at least some injection material in via one or more other injection pumps **214** and/or manifolds **216** to maintain separation of the at least some injection material. The one or more injection pipes **218** can include one injection pipe **218** or a plurality of injection pipe **218**. The one or more injection pipes **218** can be configured identically and/or have one or more variations, such as with size, dimension, or orifices differences. The one or more injection pipes **218** can be configured to output the same injection material or output different injection material among the one or more injection pipes **218**. Furthermore, in some embodiments, flow rate sensors are disposed at any position in the plumbing lines **220** to directly measure flow rate throughput. For instance, one flow rate sensor can be positioned in plumbing in-line with a first reservoir **206** and a second flow rate sensor can be positioned in-line with an injection pump **214**. In such embodiments, the one or more processors **202** can obtain flow rate signals associated with the flow rate sensor for determining and/or monitoring flow rate of an injection material at one or more locations in the plumbing lines **220**. The plumbing lines **220** can consist of aluminum, braided steel, copper, polyurethane, or other metal, plastic, synthetic, composite, or rubber material. The conductors **222** can consist of power, ground, analog, and/or data wires composed of copper, tin, gold, composite, or other conducting material, including single and multi-conductor cables. Additionally, in certain embodiments, a user interface is provided that is configured to enable interaction with the one or more processors **202** or any other component of the system **200**. The user interface can include a local user interface that is hardwired to the system **200**, such as a touch screen display and/or keypad, or can include a wireless user interface, such as an smartphone, tablet, and/or remote computer system. The system **200** can include a network, beacon, or wireless interface, such as BLUETOOTH or WIFI.

[0040] For example, system **200** can store species of Pearl Oyster, Turkey Tail, Phoenix Oyster, King Stropharia; media of liquid, sawdust, seeds, grass, enzymes, sugar; and/or gases or gaseous solutions in one or more of the reservoirs **206**. The dosing pumps **204** are usable to transfer any of the stored ingredients between reservoirs **206**, such as for mixing to create inoculation mixtures. Such inoculation mixtures may include, for example, a liquid culture of aerobic fungal decomposers, myceliated sawdust of aerobic fungal decomposers, inoculated seeds, encapsulated enzymes, or the like. The inoculation mixture in one or more reservoirs **206** is deliverable under pressure via the one or more plumbing lines **220** via the one or more injection pipes **218**. A fracture media stored in a separate reservoir **206** can be delivered under higher pressure prior to the inoculation mixture, such as using one or more injection pumps **214**. The injection pump **214** can be variably adjusted to one or more different pressure settings or a plurality of different injection pumps **214** can be employed to generate different pressures for the inoculation mixture and/or fracture media.

[0041] FIGS. **3-4** are perspective views of a device for mycoremediation, in accordance with an embodiment of the invention. APPENDIX B includes pictures of various embodiments of the device. In one instance, a device **300** for in-situ mycoremediation includes, but is not limited to, a rod casing **312** having a top end **301**, a bottom end **302**, and a sidewall **303** with one or more perforations **304**, the sidewall defining an internal channel **401** that extends from an intake opening **402** (FIG. **4**) on the top end **301** to the one or more perforations **304**; a sleeve **306** that extends around at least part of the rod casing **312** and that is slidable **307** between at least a first position **303** that covers the one or more perforations **304** and a second position **308** that at least partly uncovers the one or more perforations **304**; and a plumbing line **114** (FIG. **1**) linked to the intake opening and configured to facilitate forcible injection of one more fungal mixtures and/or air via the one or more perforations when the sleeve **306** is in the second position **308**. In certain embodiments, the rod casing **312** is cylindrical with a tapered tip **310** defining the bottom end **302**. In a further embodiment, the sidewall **303** includes a recessed segment **311** where the one or more perforations **304** are located. In a further particular embodiment, wherein the sleeve **306** is disposed within a recessed segment **311** of the sidewall **303**. In yet another embodiment, the sleeve **306** includes a detent friction lock to maintain the sleeve **306** in the first position **309** and/or the second position **308**. In one embodiment, the sleeve **306** is friction mounted within a recessed segment **311** of the sidewall **303**. In another embodiment, the sleeve is electromagnetically movable between the first position **309** and/or the second position **308**. In certain embodiments, the sleeve **306** is configured to be positioned at a point between the first position **309** and the second position **308** to partially expose the one or more perforations **304**. In yet another embodiment, the sleeve **306** is friction pushed to the first position **309** closer to the bottom end **302** of the rod **312** when the rod **312** is pushed into a subsurface (FIGS. **6** | **7**). In a further embodiment, the sleeve **306** is friction pulled to the second position **308** further from the bottom end **302** of the rod **312** when the rod **312** is retracted at least partly in the subsurface (FIGS. **6** | **7**). In certain embodiments, the one or more perforations **304** are positioned to be exposed at the second position **308** of the sleeve **306**. In some embodiments, the device **300** is configured to facilitate forcible injection of one or more fungal mixtures via the one or more perforations **304**. Also, in other embodiments, the device **300** is configured to facilitate forcible injection of air via the one or more perforations **304**. In yet a further embodiment, the sleeve **306** is slidable to one or more intermediate positions between the first position **309** and the second position **308**.

[0042] The retractable bottom-up injection tool **300** disclosed herein can by employed in direct push or CPT injection applications with high or low viscosity reagents, especially in low to medium pressure or flow applications. In one particular instance, the rod casing **312** is approximately 24-48 inches in length with an outside diameter of 1.5 inches and an inside diameter of 1 inch. The recessed segment **311** or screen of perforations **304** has an outside diameter of 1 inch and an inside diameter of ¾ inch and a length of 1 inch to multiple feet. Other sizes and shapes are within the scope of the disclosure, including different lengths, diameters, and shapes. The perforations **304** can be one or more perforations and vary in size and shape. In one particular instance, the perforations **304** are a distributed array of circular orifices in the recessed segment **311**, each approximately 1/16-¼ inch in diameter. Other shapes such as square, rectangle, triangle, oval, or other regular or irregular shape are possible. As well, other sizes are also possible, including much smaller diameter holes that are on the order of approximately 1 mm in diameter. Optionally, perforations **304** may be formed from a mesh, weave, braid, filter, screen, or other material that enables liquid or gas to permeate. The material of the rod **312** can include steel, copper, aluminum, or other metal alloy. Optionally, all or part of the rod **312** can be formed from carbon fiber, fiberglass, plastic, or other similar composite material. Furthermore, in certain instances, sleeve **306** is fixed and does not slide, thereby fixedly exposing the perforations **304**. Alternatively, sleeve **306** can be configured to rotate about threads to expose and conceal the perforations **304**.

[0043] FIG. **5** is perspective view of a device for mycoremediation, in accordance with an embodiment of the invention. APPENDIX B includes pictures of various embodiments of the device. In one instance, a device **500** for in-situ mycoremediation includes, but is not limited to, a rod casing **512** having a top end **501**, a bottom end **502**, and a sidewall **503** with one or more perforations **504**, the sidewall defining an internal channel **512** that extends from an intake opening on the top end **501** to the one or more perforations **504**; a sleeve **506** that extends around at least part of the rod casing **512** and that is slidable **507** between at least a first position **503** that covers the one or more perforations **504** and a second position **508** that at least partly uncovers the one or more perforations **504**; and a plumbing line **114** (FIG. **1**) linked to the intake opening and configured to facilitate forcible injection of one more fungal mixtures and/or air via the one or more perforations when the sleeve **506** is in the second position **508**. In certain embodiments, the rod casing **512** is cylindrical with a tapered tip **510** defining the bottom end **502**. In a further embodiment, the sidewall **503** includes a recessed segment **511** where the one or more perforations **504** are located. In a further particular embodiment, wherein the sleeve **506** is disposed within a recessed segment **511** of the sidewall **503**. In yet another embodiment, the sleeve **506** includes a detent friction lock to maintain the sleeve **506** in the first position **509** and/or the second position **508**. In one embodiment, the sleeve **506** is friction mounted within a recessed segment **511** of the sidewall **503**. In another embodiment, the sleeve **506** is electromagnetically movable between the first position **509** and/or the second position **508**. In certain embodiments, the sleeve **506** is configured to be positioned at a point between the first position **509** and the second position **508** to partially expose the one or more perforations **504**. In yet another embodiment, the sleeve **506** is friction pushed to the first position **509** closer to the bottom end **502** of the rod **512** when the rod **512** is pushed into a subsurface (FIGS. **6** | **7**). In a further embodiment, the sleeve **506** is friction pulled to the second position **508** further from the bottom end **502** of the rod **512** when the rod **512** is retracted at least partly in the subsurface (FIGS. **6** | **7**). In certain embodiments, the one or more perforations **504** are positioned to be exposed at the second position **508** of the sleeve **506**, wherein the second position **508** is closer to the top end **501** of the rod **512**. In some embodiments, the device **500** is configured to facilitate forcible injection of one or more fungal mixtures via the one or more perforations **504**. Also, in other embodiments, the device **500** is configured to facilitate forcible injection of air via the one or more perforations **504**. In yet a further embodiment, the sleeve **506** is slidable to one or more intermediate positions between the first position **512** and the second position **508**.

[0044] The retractable injection tool **500** can be employed in direct push or CPT injection applications with high or low viscosity reagents using bottom up injection, especially in high pressure or flow applications. In one particular embodiment, the rod casing **512** has a length of 48 inches, an outside diameter of 2.25 inches, and an inside diameter of 1.75 inches. Other sizes and shapes are within the scope of the disclosure. The injection ports or perforations **504** include four windows each of 0.5×1 inch orifice size and each with a conical or tear drop shape. Other shapes such as square, rectangle, triangle, oval, or other regular or irregular shape are possible. As well, other sizes are also possible, including much smaller diameter holes that are on the order of approximately 1 mm in diameter. Optionally, perforations **304** may be formed from a mesh, weave, braid, filter, screen, or other material that enables liquid or gas to permeate. The material of the rod **312** can include steel, copper, aluminum, or other metal alloy. Optionally, all or part of the rod **312** can be formed from carbon fiber, fiberglass, plastic, or other similar composite material. Furthermore, in certain instances, sleeve **506** is fixed and does not slide, thereby fixedly exposing the perforations **504**. Alternatively, sleeve **506** can be configured to rotate about threads to expose and conceal the perforations **504**.

[0045] In certain embodiments, the injection remediation system **100** is configured to utilize a plurality of injection pipes **112** that include one or more of the device **300** and/or device **500**. For example, the injection remediation system **100** can include a plurality of device **300** and a plurality of device **500** that are intermixed, but separately plumbed to one or more different pressurized plumbing lines **114** and/or sourced from one or more different reservoirs **102**. Thus, at one injection mycoremediation site **101**, the device **500** may be employed first to inject higher pressure fracture medium and the device **30** may be employed second to inject lower pressure fungi and/or mycelium, oxygen, or air. Thus, a plurality of different device **300** and device **500** can be used in parallel or sequentially to deliver different media at different pressures efficiently during a remediation period.

[0046] FIGS. **6-8** are environmental perspective views of a device for mycoremediation, in accordance with an embodiment of the invention. In one instance, the injection head **300** is driven into a site **101** for mycoremediation. The device **300** has at least two configurations of a first position **309** where the perforations are covered and a second position **308** where the perforations are uncovered and exposed to the site **101**. Pushing the top end **301** of the device **300** downward into the site **101** collapses the device **312** into the first position **309**. Pulling or retracting the device **300** at least partly from the site **101** using the top end **301** extends the device **312** into the second position **308**. Alternatively, using a threaded embodiment, rotating the device **300** at the top end **301** in one direction can conceal the perforations **304** in the first position **309**. Oppositely, rotating the device **300** at the top end **301** in an opposing direction can expose the perforations **304** in the second position **308**. The degree to which the perforations **304** are exposed, or the partial position of the device **300** between the first position **309** and the second position **308** can be controlled based upon an extent of pulling force, rotational force, or, in certain instances, using one or more of a stopper pin, retainer pin, spacer, or locking nut. The device **300** is illustrated as being partially inserted into the site **101**, but the device **300** can be driven or forced to various depths in order to position the perforations **304** deeper or shallower within the site **101**. For instance, the device **300** can be driven fully below and into the site **101** such that the top end **301** is completely below a top surface of the site **101**. In certain embodiments, the device **300** can be stepped through different depths sequentially, to fracture and/or inject various mycoremediation material within different depth planes of the site **101**. A plurality of devices **300** can be used to create a 3D matrix of fracture/injection locations at different X, Y, and Z coordinates.

[0047] In one embodiment, as depicted in FIG. **8**, a plurality of injection pipes **300** are driven or vibrated into the ground site **101** with a “direct push” type rig. As the zone of interest within the site **101** is reached, the rods **300** are retracted slightly (e.g. 1 to 18 inches), which exposes a series of injection slots in the drive pipe **300**. A propagation notch or fracture **802** is started using pressurized water. This notch **802** is about 4 to 6-inches deep into the lateral soil horizon. An inoculation fluid is then pumped into the pipe **300** under pressure until the overburden pressure of the soil in site **101** is exceeded and the horizon fractures. The lateral fracture then extends out horizontally and is approximately 1-inch wide and has approximately a 30-degree radial expression. The tooling **300** has the capability to inject both aqueous and viscous slurry solutions, such as those including grain/sawdust material and mycelium as a slurry. In certain embodiments, tooling **300** allows for higher hydraulic pressure to help create fractures in low porosity soils **101** and allows for lower hydraulic pressure to support injection of mycelium and/or enzymes at a lower sustained pressure.

[0048] The devices **300** therefore enable efficient robust growth of fungi in a hyphal network **803** for mycoremediation. That is, the devices **300**, when used in aquatic or terrestrial environments **101**, soil porosity, permeability, and hydraulic conductivity are first increased by fracturing a subsurface **101** with a first fluid, such as a carbon and/or water solution, at a high pressure. Then, a slurry of fungi with a food source is disbursed into the disturbed subsurface under a pressure that is lower than that of the first fluid to initiate and support colonization and/or formation of a matrix of mycelium **803**. The matrix of mycelium itself increases soil porosity and produces extracellular ligninolytic enzymes that breakdown and degrade pollutants. Further, the matrix of mycelium acts to absorb and/or filter various pollutants, such as heavy metals to support remediation. Additionally, the area of influence can be increased via the addition of more sugar solution to encourage the migration of fungus. More solution may be added to the original injection pipe **300**, or via one or more additional injection pipes **300** to encourage hyphal fungal growth. Additionally, either before, after, or during any of the foregoing operations, an optional injection of oxygen or air can be forcibly injected to the subsurface or encouraged through venting. Thus, the device **300** is configured to remain in-situ with passive ventilation of air via the one or more perforations. It is also possible to restrict access to air in the subsurface for anoxic substrates, such as by capping the device **300** at the top end **301**.

[0049] FIG. **9** is a flow diagram of a process for mycoremediation, in accordance with an embodiment of the invention. In one instance, a process for in-situ mycoremediation **900** is provided, including at least driving one or more rod casings into a subsurface zone at **901**, the one or more rod casings including one or more orifices that are covered; partially retracting the one or more rod casings to expose the one or more orifices at **902**; forcible injecting a fluid via the one or more orifices to initiate fracture in the subsurface zone at **903**; forcibly injecting a fungal nutrient mixture via the one or more orifices into the subsurface zone at **904**; and ventilating the subsurface zone with air via the one or more orifices to support a hyphal network of mycelium in the subsurface zone at **905**, which in some cases can include pressurized air.

[0050] In one embodiment, a liquid fungal mixture can be established as follows, in small batch as described or in large batch with substantially equivalent ratios and steps. Mix approximately 500 mL of non-chlorinated water, approximately 2 Tbsp of dextrose, and approximately 2 Tbsp of malt extract over low heat to dissolve. Pressure cook the mixture at approximately 15 PSI for approximately 20 minutes. Inoculate the mixture with fungi and place inoculated mixture in a dark environment with stable room temperature. Aerate for approximately 30 minutes daily for approximately 2-3 weeks. Refrigerate the liquid mycelium rich fungal mixture until use.

[0051] In certain embodiments, the liquid nutrient mycelium mixture is established next by adding a carbonous substrate to the liquid mycelium mixture to enable the fungal mycelium to do remediation work through decomposition and enzymatic production. In one particular embodiment, sawdust is used as the carbonous substrate and sawdust spawn is created as follows, in small batch as described or in large batch using the same approximate ratios and steps. First, combine hardwood (alder) sawdust and with ½ cup wheat or rice bran and ¼ cup gypsum to jump start nutrition. Hydrate each separately or together. Soak sawdust in moisture for a few hours to overnight. Drain the sawdust mixture such that it is moist but not wet. Pack the sawdust mixture into an airtight sterile container. Then, inoculate the sawdust mixture with the mycelium mixture using an approximate 10% inoculation ratio (1:10 spawn to substrate).

[0052] The nutrient mycelium mixture is usable for injection via the devices and systems described herein. However, many other methods of producing fungal mixtures are possible and usable with the present invention for injection remediation.

[0053] The present invention may have additional embodiments, may be practiced without one or more of the details described for any particular described embodiment, or may have any detail described for one particular embodiment practiced with any other detail described for another embodiment. For instance, the hyphal network can alternatively be used to increase the hydraulic conductivity of the soil by creating pathways through interstitial grains. The grains can be “locked” together with a precipitate then can stabilize the soil to mitigate spread failure, circular failure, or prevent water erosion. Once the soil has a lower hydraulic conductivity, different mixtures of microbial fluid or inoculant can then optionally be added to address any subject condition.

[0054] While preferred and alternate embodiments of the invention have been illustrated and described, as noted above, many changes can be made without departing from the spirit and scope of the invention. Accordingly, the scope of the invention is not limited by the disclosure of these preferred and alternate embodiments. Instead, the invention should be determined entirely by reference to the claims that follow.

Claims

1-20. (canceled)
21. A injection device for remediation, the injection device comprising: a casing including one or more perforations; a sleeve that extends around at least part of the casing, the sleeve being movable between at least a first position that covers the one or more perforations and at least one second position that at least partly uncovers the one or more perforations, wherein the injection device is configured to inject one more remediation materials into a subsurface via the one or more perforations.
22. The injection device of claim 21, wherein the casing comprises a top end with an intake opening, a bottom end with a tip, and a sidewall, the sidewall defining an internal channel that extends from the intake opening to the one or more perforations.
23. The injection device of claim 21, wherein the casing further comprises a recessed segment, the one or more perforations being located within the recessed segment.
24. The injection device of claim 21, wherein the one or more perforations are one or more of the following types: orifice, mesh, filter, or screen.
25. The injection device of claim 21, wherein the one or more perforations are defined by one or more of the following shapes: slot, circle, square, rectangle, triangle, tear drop, hole, or oval.
26. The injection device of claim 21, wherein the sleeve extends entirely around the casing.
27. The injection device of claim 21, wherein the sleeve extends around a recessed segment of the casing.
28. The injection device of claim 21, wherein the sleeve is slidably pushed to the first position and pulled to the at least one second position.
29. The injection device of claim 21, wherein the sleeve is rotated in one direction to the first position and in an opposite direction to the at least one second position.
30. The injection device of claim 21, wherein the sleeve is movable to one or more intermediate positions between the first position and the at least one second position.
31. The injection device of claim 21, wherein the first position or the at least one second position are defined by one or more of a stopper pin, spacer, or locking nut.
32. The injection device of claim 21, wherein the injection device is configured to forcibly inject one more remediation materials into a subsurface via the one or more perforations.
33. The injection device of claim 21, wherein the injection device is configured to inject one or more of the following mediums: fluid, slurry, or gas.
34. The injection device of claim 21, wherein the injection device is configured to inject one or more of the following types of remediation materials: fungus, mycelium, enzyme, bacteria, or microbe.
35. The injection device of claim 21, wherein the injection device is configured to inject one or more gaseous remediation materials.
36. The injection device of claim 21, further comprising: a plumbing line linked to an intake opening of the casing; and a pump, wherein the pump pressurizes the one or more remedial materials via the plumbing line for output via the one or more perforations.
37. A injection device for remediation, the device comprising: a casing including at least: a top end with an intake opening, a bottom end with a tip, a recessed segment with one or more perforations, a sleeve around the recessed segment, and a sidewall defining an internal channel that extends from the intake opening to the one or more perforations, wherein the sleeve is movable between a first position that covers the one or more perforations and a second position that at least partly uncovers the one or more perforations.
38. A device for in-situ remediation, the device comprising: a casing including one or more perforations for dispensing one or more remediation materials; and a sleeve disposed around the

casing and that rotates between a first position covering the one or more perforations and a second position exposing the one or more perforations.